

Solid state NMR structural studies of the lithiation of nano-silicon: Effects of charging capacities, host-doping, and thermal treatment

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ABSTRACT

The electrochemical lithiation of undoped, P-doped and B-doped nano-silicon particles (100–200 nm diameter) has been studied during the first cycle by ex-situ ⁶Li and ⁷Li magic angle spinning nuclear magnetic resonance (MAS-NMR) spectroscopy. Samples were charged within pouch cells up to capacities of 4000 mAh/g, at C/40 followed by NMR analysis. The spectra reveal important quantitative information on the local lithium environments during the various stages of the intercalation process. Approximate Li/Si ratios of the lithium silicides present in the nanoparticles can be deduced, based on the initial formation of the SEI layer, which accounts for an irreversible capacity of up to 500 mAh/g. Surface lithium silicide environments with high Li concentrations (corresponding to the composition Li₁₅Si₄) are preferentially formed at charging capacities near 1000 mAh/g. At higher charging capacities, irreversible capacity losses are lower and a wide distribution of lithium silicide environments is found, resembling those present in the crystalline phases Li₁₂Si₇, Li₇Si₃, and Li₁₃Si₄. At a charging capacity of and above 2000 mAh/g the large majority of silicon is converted to lithiated silicide particles. Boron-doped nano-Si materials behave generally similar, while phosphorus-doping reveals clear beneficial effects, in particular concerning the initial lithiation stages. Both irreversible capacity losses and surface “over-lithiation” are significantly diminished in these samples. Exposure of lithiated nano-Si samples to elevated temperatures (400–440 K) results in the crystallization of Li₇Si₃ in all of those nano-Si samples charged with at least 1500 mAh/g.

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1. Introduction

Lithium-ion batteries are attracting a lot of interest in academic and industrial research for realizing energy storage devices for portable consumer electronics [1] as well as electro-mobility and stationary electricity [2]. The limitations of current Li-ion battery components, in terms of capacity, reversibility and durability, for this application prospect demand a search for new suitable active materials, both on the cathode and on the anode side [3–5]. Nano-silicon is one of the most promising potential anode materials, owing to its larger theoretical charge capacity (3579 vs. 372 mAh/g of graphite) [5–9]. However, the dramatic volume expansion during lithiation (about 300%) encountered with this material poses a huge engineering challenge before this material can be routinely used as an active material [6,7,10–13].

During the first lithiation cycle, crystalline nano Si undergoes a progressive amorphization [14], as also verified by in situ X-ray diffraction studies [15]. To elucidate the lithiation mechanism structural information is essential, which is, however, difficult to obtain by X-ray powder

diffraction for amorphous samples. Recently ⁶Li and ⁷Li solid state nuclear magnetic resonance (NMR) techniques have been applied successfully to this problem [16–18]. These studies have generally detected multiple Li species interacting with a range of different Si-environments. According to theoretical work during the early stages of the first lithiation cycle the Si-lattice is broken down into interconnected Si-chains and rings consisting of 5 to 8 Si atoms. Increasing the Li concentration, these structures are broken down into smaller Si₄ and Si₃ clusters, which are eventually transformed into Si₂ dumbbells and, finally, isolated silicon species [19,20]. Fully lithiated silicon corresponds to a Li_{3.75}Si phase, which crystallizes to Li₁₅Si₄ if the voltage is lower than 50 mV (vs Li/Li⁺) [21]. NMR studies of crystalline lithium silicide phases (Li₁₂Si₇, Li₇Si₃, Li₁₃Si₄, Li₁₅Si₄, and Li₂₁Si₅) have shown that the ⁷Li chemical shifts are quite sensitive to the different types of silicon species they interact with [16]. This is true even though the spectra observed at room temperature are typically dynamically averaged. More structural and dynamic details have been recently resolved on some crystalline lithium silicides using variable-temperature ⁶Li and ²⁹Si MAS-NMR [22–25], even though the lithium spectra are still affected by ionic motion within the accessible temperature range (T ≥ 200 K). While in the case of amorphous Li_xSi_y materials recent results have suggested that the local lithium environments are generally comparable to those found in the

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crystalline lithium silicides [18], specifying detailed phase compositions based on the NMR results can only be considered approximate. Still, a distinction of different types of lithium ions is possible on the basis of the individual chemical shift ranges observed for different lithium species. For example, Key et al. in their ^7Li MAS-NMR study have made the distinction between Li-ions near silicide clusters (resonating around 18 ppm), and lithium ions interacting predominantly with isolated silicon anions (resonating around 6 ppm) [16,17]. Despite the rather long spin-lattice relaxation times and lower natural abundance of ^6Li , the NMR observation of this nucleus is often preferable to ^7Li , because of the improved spectral resolution available. ^6Li MAS-NMR spectra are virtually unaffected by homonuclear dipolar broadening and also lack the spinning sidebands generated by first order quadrupolar perturbation effects [18]. In the present contribution we explore the use of ^6Li NMR to analyze the lithium local environments of amorphous lithiated nano-silicon as a function of charging state. In this connection, we also explore the effect of boron- and phosphorus-doping, as it was previously reported that the addition of dopants to Si influences the kinetics of Li insertion, enhancing the cycle performance in the case of P-doping [26–29]. Furthermore, thermally induced crystallization phenomena of these amorphous lithiated phases are monitored by ex-situ ^{29}Si as well as variable temperature ^7Li solid state NMR experiments, and compared with previous DSC results [30,31].

2. Experimental

2.1. Nano-silicon preparation and doping

The silicon nanoparticles were prepared by microwave induced electrical discharge dissociation of gases in a low pressure flow reactor. For the n-doped silicon particles silane and phosphine (dopant concentration $1.22 \times 10^{18}/\text{cm}^3$) were used while silane and borane (dopant concentration $1.30 \times 10^{19}/\text{cm}^3$) were mixed for the preparation of the p-doped silicon particles. Electron micrographs indicate average particle sizes of 37 and 56 nm for boron- and phosphorus-doped nanosilicon samples, respectively (see Figures S1 and S2, Supplemental Materials).

2.2. Electrode preparation

For the electrode preparation a mixture of 10 wt% carbon black C65 (Timcal), 80 wt% pure or doped nano-silicon, and 10 wt% of Na-CMC (Dow Wolff Cellulosics) dissolved in water were dispersed using a bead mill (Dispermat LC-2 VMA-GETZMANN GMBH, Germany). The obtained slurry was cast on a copper current collector, with a wet film thickness of 100 μm . The electrode films were pre-dried at 80 °C under ambient atmosphere (30 minutes) and further dried at 120 °C under vacuum over night.

2.3. Cell assembly and charge

The anodes were cut into $5 \times 5 \text{ cm}^2$ large squares. Pouch cells were assembled from these squares, commercially available lithium iron phosphate cathodes ($5.2 \text{ cm} \times 5.2 \text{ cm}$, Litarion™, Evonik GmbH, Germany) and Freudenberg FS 2620 (Freudenberg Vliesstoffe KG, Germany) as separator. 1 M LiPF_6 in ethylene carbonate/diethyl carbonate EC/DEC = 3:7 by weight (UBE, Japan) was used as electrolyte. The cell was sealed under vacuum. The pouch cells were charged on Maccor S4000 Battery Tester (MACCOR Ltd., USA) with constant current (CC) of C/40 until the desired specific capacity for each cell was reached. For each of the silicon electrodes, six pouch cells were prepared for the MAS-NMR analysis corresponding to charge capacity levels of 500, 1000, 1500, 2000, 3000, and 4000 mAh/g, respectively (denoted samples C-500, C-1000, etc. in the following). The final charging level of 4000 mAh/g takes into consideration that up to 500 mAh/g may be consumed during the process of solid/electrolyte interface (SEI) formation.

2.4. Solid state NMR

The dried anodes were scratched from the Cu current collector foil, and after grinding in an agate mortar under an Ar atmosphere in a glove box, the powder was packed in 4 mm rotors for MAS-NMR. The B- and P-doped samples were diluted with dried boron nitride in a 1:1 mass ratio in order to get a complete filling of the rotor. All these operations were performed in a glovebox under dried Argon flux (levels of O_2 and $\text{H}_2\text{O} < 1 \text{ ppm}$). $^{6,7}\text{Li}$ and ^{29}Si MAS-NMR studies were carried out on a Bruker DSX 400 spectrometer using either a 4 mm Bruker WTN probe or a 4 mm Bruker VT MAS triple resonance probe. Dry nitrogen gas was used for variable temperature MAS experiments with typical spinning speeds of 8.0 kHz. Room temperature measurements were performed with spinning speeds ranging from 8.0 to 12.0 kHz. Differences between set and actual temperature were calibrated using the ^{207}Pb resonance of lead nitrate as previously described [32,33]. ^7Li NMR spectra were acquired at 155.56 MHz using 90° pulses of 3.2 to 6.5 μs length and a recycle delay between 10 and 40 s (values optimized on each individual sample). ^6Li NMR spectra were acquired at 58.09 MHz with 90° pulses of 5.8–8.0 μs length and a recycle delay ranging from 60 to 120 s (optimized on each sample). $^6\text{Li}\{^7\text{Li}\}$ CP MAS experiments were performed after the optimization of the cross polarization conditions with Li_2SiO_3 . A B_1 -amplitude corresponding to a nutation frequency of 35.7 kHz, a ^7Li -90° pulse of 5.0 μs , a contact time of 3.5 ms, and a recycle delay of 5 s were used. Chemical shifts are reported relative to a 1 M aqueous solution of LiCl. ^{29}Si MAS-NMR spectra were acquired at 79.5 MHz at a spinning speed of 8.0 kHz. For anode samples a 90° pulse of 3.0–4.5 μs length and a recycle delay of 10 s were used. For pure nano-Si a 90° pulse of 6.0 μs length and a recycle delay of 80 s were used. Chemical shifts are reported relative to TMS.

3. Results and discussion

3.1. Effect of charging capacity

Fig. 1 shows three different series of ^7Li MAS-NMR spectra obtained at different charging capacities for each active material. As previously observed [16–18], the lithiation process for pure nano-Si anodes starts first with the formation of the solid/electrolyte interface (SEI) layer (signal at about 0 ppm), followed by the generation of lithium in different amorphous silicide environments, resonating between 3 and 21 ppm. At the maximum lithium loading (sample C-4000), the majority of lithium is present in the form of a $\text{Li}_{15}\text{Si}_4$ -like phase at voltages lower than 50 mV (vs Li/Li^+). While this is the case for the samples of all three series studied, some important differences in the lithiation product distributions are evident, which may be attributed to the presence of dopants in the starting active electrode material.

For more detailed structural insights, the ^6Li MAS-NMR spectra, summarized in Fig. 2, offer improved spectroscopic resolution. Even though the Si-environments explored by Li in the amorphous nano-phases generated during the electrochemical lithiation of nano-silicon may differ somewhat from those found in the known crystalline lithium silicides, the NMR shifts of the latter can be used to guide the peak deconvolution in terms of several distinct components (see also Table 1):

- signals in the range 16–22 ppm are attributed to Li atoms in cluster-rich environments, where Si_5 rings and Si_4 stars can co-exist, as they do in crystalline $\text{Li}_{12}\text{Si}_7$ (theoretical capacity 1635 mAh/g),
- signals in the range 12–14 ppm are attributed to lithium species interacting with smaller fragments, down to the Si_2 dimers present in crystalline Li_7Si_3 (2226 mAh/g),
- signals in the 8–10 ppm range describe lithium environments interacting with both dimeric and isolated Si (in analogy to the situation in crystalline $\text{Li}_{13}\text{Si}_4$, 3101 mAh/g),

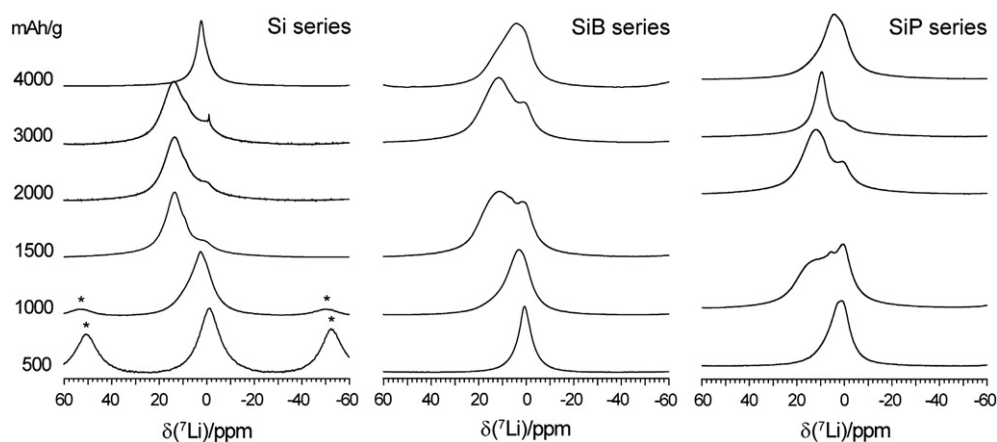


Fig. 1. ^7Li MAS-NMR spectra recorded at room temperature for the anodes of the undoped, boron-doped and phosphorus-doped nano-Si samples lithiated from 500 to 4000 mAh/g. Spinning sidebands are indicated by asterisks.

- d) signals in the range 3–6 ppm correspond to Li species interacting exclusively with isolated Si anions (corresponding to the situation in $\text{Li}_{15}\text{Si}_4$, 3578 mAh/g),
- e) signals in the range –1 to 3 ppm can be attributed to salt-like lithium species present at the SEI [34,35]. The SEI arises from decomposition/oxidation processes and/or residual adherent electrolyte material.

Peak deconvolutions of the ^6Li MAS-NMR spectra were carried out with the DMFIT program [36], minimizing variations of the positions and widths of the individual spectral components between different samples. In certain cases, lithium species of a given kind are represented by more than one signal component within the corresponding chemical shift range. This is particularly true for the signal characteristics of $\text{Li}_{12}\text{Si}_7$ -like environments, which are frequently described by two distinct components near 17 and 22 ppm. Table 1 summarizes the quantitative deconvolution results. The quantifications of the various lithium species done in this manner can be used further to deduce the amount of reacted nano-silicon in each sample, as described in reference [18]. This approach is based on the working, albeit approximate, assumption that the local environments identified above resemble the lithium silicide stoichiometries of the known crystalline phases. For example, at a

charging capacity of 1000 mAh/g, 1 gram, i.e. 0.0356 mole of Si have been exposed to 3.6×10^3 C, i.e. 2.25×10^{22} elemental charges = 0.0374 mole of Li. From the NMR lineshape analysis we find that 48%, i.e. 0.0179 mole end up as electrochemically inactive, salt-like species, 36%, i.e. 0.0135 mole are found in a $\text{Li}_{15}\text{Si}_4$ -like environment, and 16%, i.e. 0.006 moles are present in a $\text{Li}_{13}\text{Si}_4$ -like form. Considering the different Li/Si ratios of these two compositions, we can thus deduce that each gram of electrode material contains 0.0036 and 0.0018 moles of silicon present in $\text{Li}_{15}\text{Si}_4$ - and $\text{Li}_{13}\text{Si}_4$ -like forms, respectively. Altogether, 0.0054 moles of Si per gram of electrode, i.e. 15%, are present in the form of lithium silicides, the remainder being either unreacted silicon or possibly lithium silicate formed by corrosion of the lithium silicide. Similar calculations can be done for all the other samples, and the results are summarized in Table 2.

3.2. Undoped nano-Si

Turning first to the spectra of the undoped Si samples, relatively large irreversible capacity losses are observed at low charging capacities (samples C-500 and C-1000). Thus, the sample C-500 only shows a single peak near –0.7 ppm, indicating that the entire amount of lithium

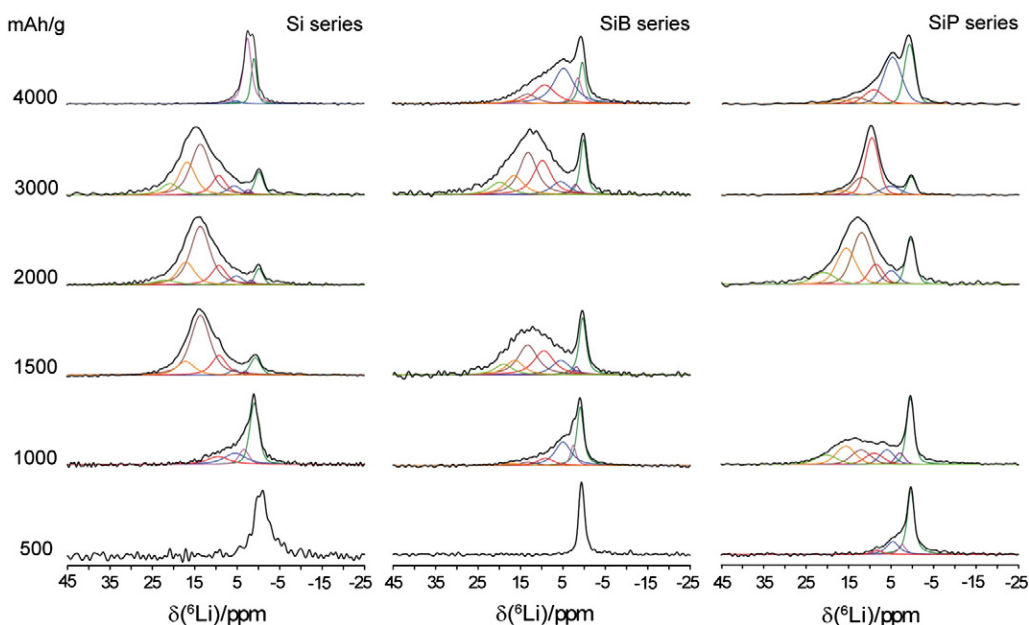


Fig. 2. ^6Li MAS-NMR spectra recorded at room temperature for the anodes of undoped, boron-doped and phosphorous-doped nano-Si samples lithiated from 500 to 4000 mAh/g. The colored lines are line-shape deconvolution components.

Table 1
⁶Li NMR isotropic chemical shifts (± 0.5 ppm), peak assignments and quantification of the different lithium species based on area percentages ($A \pm 2\%$) obtained via line-shape deconvolution (Fig. 2).

	500 mAh/g		1000 mAh/g		1500 mAh/g		2000 mAh/g		3000 mAh/g		4000 mAh/g	
	δ_{iso} (ppm)	A (%)	δ_{iso} (ppm)	A (%)	δ_{iso} (ppm)	A (%)	δ_{iso} (ppm)	A (%)	δ_{iso} (ppm)	A (%)	δ_{iso} (ppm)	A (%)
<i>nano-Si</i>												
SEI	−0.7	100	1.0	48	0.6,3.1	10	−0.1,1.7	7	−0.2,2.4	10	0.9	29
Li ₁₅ Si ₄	–	–	–	–	–	–	–	–	–	–	2.4	67
isolated Si	–	–	3.4, 5.5	36	5.6	3	5.2	6	5.7	6	5.2	4
iso.Si/dimers	–	–	9.5	16	9.2	16	9.3	14	9.3	12	–	–
dimers	–	–	–	–	13.6	59	13.7	51	13.7	40	–	–
clusters	–	–	–	–	17.0	12	17.2	22	17.2	32	–	–
<i>B-doped Si</i>												
SEI	0.6	100	0.9	36	0.4,1.8	21			0.2,2.0	17	0.3,1.4	26
Li ₁₅ Si ₄	–	–	–	–	–	–			–	–	–	–
isolated Si	–	–	2.6,5.0	45	5.5	12			5.6	9	4.7	39
iso.Si/dimers	–	–	9.1	11	9.5	21			9.8	23	9.2	22
dimers	–	–	13.2	5	13.3	26			13.2	29	13.2	10
clusters	–	–	17.0	3	16.2	20			16.2	22	16	3
<i>P-doped Si</i>												
SEI	0.4	63	0.4,2.9	36			0.5	16	0.4	10	0.8	33
Li ₁₅ Si ₄	–	–	–	–			–	–	–	–	–	–
isolated Si	2.4,4.6	31	6.0	11			5.1	6	5.3	11	4.9	43
iso.Si/dimers	8.3	6	9.0	10			8.7	10	9.7	51	9.3	14
dimers	–	–	12.1	14			12.1	35	12.0	22	13.2	6
clusters	–	–	16.2	29			16.2	33	17.0	6	17.0	4

electrochemically provided reacted to form salt-like decomposition products, such as LiF, Li₂O, Li₂CO₃ and lithium silicates. In sample C-1000, still roughly half of the electrochemically provided lithium is again lost to this process, whereas the other half is found in a local environment characterized by the interaction with monomeric silicon species. This environment, which corresponds to a Li₁₅Si₄-like phase, actually represents a lithiation product that is expected at much higher Li/Si ratios than that corresponding to 1000 mAh/g. Its formation at this early lithiation stage suggests a very inhomogeneous distribution of the lithium within the lithiated particles, consisting of a shell of “over-lithiated” lithium silicide and a core of un-reacted silicon. Indeed, the quantitative NMR peak deconvolution indicates that only 15% of all silicon is present in the form of a lithium silicide in this sample.

The decomposition products detected in the ⁶Li NMR spectra near 0 ppm form part of an “interphase” at the interface between the solid anode and the electrolyte [37], which eventually protects the lithium silicides formed at the anode from corrosion by the electrolyte material during cell cycling. During the first cycle at low charging capacities, this SEI may not yet be fully developed, particularly considering the large volume expansion associated with the transformation from nanocrystalline silicon to the highly “overcharged” Li₁₅Si₄-like material. Thus, a major factor contributing to the large amount of lithium lost to these salt-like decomposition products may be the corrosion of these silicide particles by the electrolyte, with which they remain in contact during the waiting time between the electrochemical lithiation and the ex-situ NMR measurement.

As lithiation progresses through the 1500, 2000 and 3000 mAh/g stages (samples C-1500, C-2000, and C-3000), significant changes are

observed in the spectra. In this regime, about 90% of the Li atoms offered electrochemically are being intercalated into the nano-silicon. Secondly, the spectra and Table 1 indicate that the dominant ⁶Li NMR peak is observed near 13.6 ppm, corresponding to lithium ions interacting with silicon dimers and possibly other small fragments. In addition, substantial amounts of lithium species also interact with larger clusters such as present in Li₁₂Si₇, as suggested by the observation of two distinct peaks near 16–17 and 20–22 ppm. In contrast, lithium species in Li₁₅Si₄-like environments are largely suppressed in this concentration range. Within the 1500 to 3000 mAh/g range, the spectra appear nearly independent of composition. In this region lithiation progresses under eventual conversion of the entire silicon nanoparticles to lithium silicides without major changes in the product distribution: the fraction of silicon converted into lithium silicide products as estimated by the approximate analysis described above changes from 60% to 85% to 100% at the charging capacities of 1500, 2000, and 3000 mAh/g, respectively. Also, the irreversible capacity losses, as deduced from the spectral area of the ⁶Li NMR peak near zero ppm, appear to be consistently lower than those measured at the earlier lithiation stages, corresponding only to 150–300 mAh/g. This effect appears to be a systematic one and argues in favor of the post-lithiation corrosion processes discussed above. Those Si materials that have been provided with a relatively small amount of charge, i.e., have been taken out from the electrochemical cell at the beginning of the charge process, have not been completely covered with an SEI and thus do suffer from corrosion with the electrolyte, leading to large amounts of Li to be identified in the SEI layer. In contrast, the Si materials at the charging capacities between 1500 and 3000 mAh/g are less sensitive towards corrosion because the SEI is more fully developed in these samples and hence can exercise its protective role more effectively.

Finally, in the sample corresponding to a charge capacity of 4000 mAh/g the major signal component is centered around 2.4 ppm and can be attributed to lithium in a Li₁₅Si₄-like environment. In this sample, a substantial amount (29%) of Li is found to be irreversibly lost to salt-like environments, contributing to a signal near 0.9 ppm. In addition, the analysis of the spectrum suggests that only 80% of the silicon is present in the form of the lithium silicide product. Clearly, the ⁶Li NMR spectra reveal that between 3000 and 4000 mAh/g the lithium silicide phases undergo substantial re-structuring. Such re-structuring, accompanied by further volume expansion, can also lead to SEI ruptures and facilitate further corrosion processes in the material.

Table 2
 Fraction of silicon converted to lithium silicide nanoparticles as a function of charging capacity.

Material	Charging capacity (mAh/g) →					
	C-500	C-1000	C-1500	C-2000	C-3000	C-4000
nano-Si	0%	15%	60%	85%	100%	80%
Si(B)	0%	20%	51%		100%	98%
Si(P)	5%	30%		82%	100%	87%

3.3. Effects of boron and phosphorus doping

Besides revealing the same broad general trend concerning the lithium silicide product distribution, as a function of charging capacity, Fig. 2 also shows some important differences between the undoped, the B-doped and the P-doped samples, which are again best discussed as a function of charging capacity.

At low charging capacities, the P-doped samples show significantly lower irreversible capacity losses in the spectra. Thus, at 500 mAh/g, only 63% of the Li atoms provided are lost to salt-like products, corresponding to an irreversible capacity loss of 315 mAh/g. At 1000 mAh/g, the capacity loss is 360 mAh/g both in the B-doped and the P-doped samples, which is again significantly lower than that (480 mAh/g) in the undoped sample. Furthermore, the P-doped sample shows significantly different lithiation product distributions at these low charging capacities. While in the undoped and in the B-doped samples the majority of the Li atoms interact with monomeric silicon species (peak at 5.5 ppm, corresponding to the $\text{Li}_{15}\text{Si}_4$ -like phase), the P-doped sample shows a much wider product distribution, in which all of the different possible species are represented. From the quantitative peak deconvolution we can deduce that 30% of silicon has been converted to lithium silicide products in the P-doped sample, which is twice the amount of that converted in the undoped material. Thus, it appears that P-doping can to some extent diminish the local “over-lithiation” observed in the undoped material at low charging capacities.

At a lithiation capacity of 1500–2000 mAh/g the irreversible capacity losses are found to be lowest in the undoped nano-Si materials, about 150 mAh/g instead of 320 mAh/g for the corresponding B- and P-doped samples. In the undoped sample the majority of the lithium atoms interact with Si_2 dimers, whereas in the doped materials the product

distribution is much wider, comprising all the environments. The fractions of silicon present in the form of lithium silicides are similar at these capacity values. At 3000 mAh/g the analysis shows that in all the three samples silicon has been converted quantitatively to lithium silicide phases. Still, the spectra reveal major differences in the width of the product distribution. The most homogeneous lithiation appears to have been achieved in the P-doped material. Finally, at a charge capacity of 4000 mAh/g, all three samples show large irreversible capacity losses, signifying conversion of substantial amounts of lithium to salt-like (SEI) environments. While $\text{Li}_{15}\text{Si}_4$ -like amorphous environments appear to be the major lithiation product in all three cases, the B- and P-doped samples indicate a larger diversity of lithium silicide environments present at this charging state. The slow transport kinetics associated with the $\text{Li}_{15}\text{Si}_4$ -like nanophase are probably responsible for the poor Coulombic efficiency of these electrodes. Based on the results of the present study, we postulate that better reversibility will be found if lithiation is limited to charging capacities at which $\text{Li}_{15}\text{Si}_4$ -like environments can be avoided. Our results suggest this to be the case near 3000 mAh/g.

We also tested both single pulse ^7Li , ^6Li as well as $^6\text{Li}\{^7\text{Li}\}$ CP MAS-NMR at temperatures down to 200 K. As illustrated in Figure S3 and S4, lowering the temperature in the present amorphous samples results in a significant loss of resolution. This situation contrasts that in crystalline lithium silicides, where substantially increased site resolution can be obtained at sufficiently low temperature, where dynamic averaging processes are frozen out [22,25]. Evidently, in the present amorphous samples the lithium species detected at low temperatures are subject to a wide chemical shift distribution, resulting in poorly resolved spectra with strongly overlapping contributions. In contrast, fast lithium motion on the MAS-NMR timescale leads to dynamical averaging phenomena at

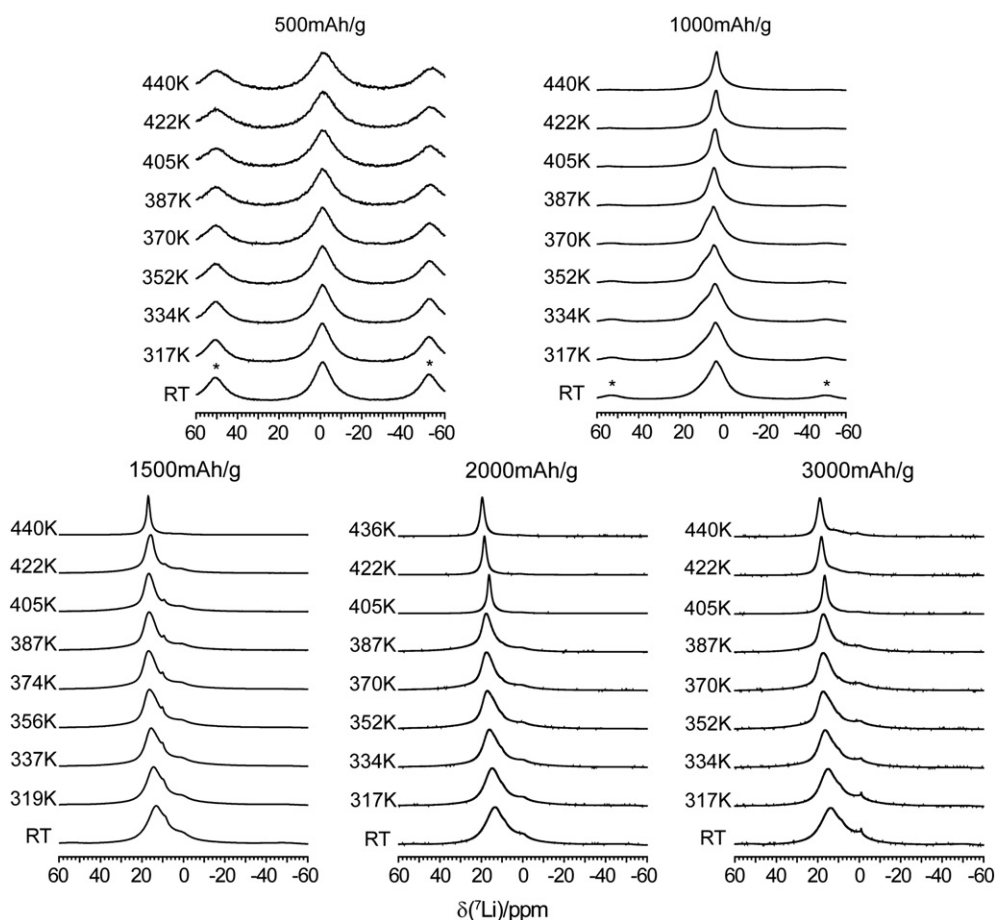


Fig. 3. ^7Li MAS-NMR spectra recorded at elevated temperatures for several lithiated nano-Si anodes. Spinning sidebands are indicated by asterisks.

room temperature, allowing a quantitative differentiation of distinct signal contributions. While at room temperature intra-particle dynamic averaging appears to be fast on the MAS-NMR timescale of 10^{-2} seconds, inter-particles transfer is slow, resulting in separate spectra for separate nanoparticles, reflecting different lithiation stages within a given sample.

3.4. MAS-NMR measurements at elevated temperatures and on heated samples

Fig. 3 shows the ^7Li MAS-NMR spectra for undoped silicon at five different lithiation stages recorded in the temperature range 297 K to 440 K. Some selected spectra from this series are shown in Fig. 4, which also displays the room temperature spectra of these samples subsequent to annealing. At the charging capacity of 500 mAh/g no changes can be observed. Evidently, the salt-like species, which constitute the SEI layer, are thermally stable and no decomposition takes place. Strong spinning sidebands reflecting the anisotropy of the nuclear electric quadrupolar broadening of the ^7Li non-central Zeeman transitions indicate that these lithium species continue to be immobile over the entire temperature range investigated. At the charging capacity of 1000 mAh/g about 50% of the lithium is inserted in the silicon particles, mostly in a bonding environment dominated by monomeric silicon species. Increasing the temperature, we observe a progressive line narrowing leading at 440 K to a sharp peak centered at 2.6 ppm. Returning stepwise to room temperature, the peak broadens again, indicating full reversibility. For the C-1500 sample the thermal treatment indicates a progressive narrowing of the shoulder at 9.5 ppm, attributed to lithium

in a $\text{Li}_{13}\text{Si}_4$ -like environment containing both monomeric Si and Si_2 dumbbells. As the temperature increases beyond 300 K, this peak shows progressively reduced intensity, while the main resonance previously centered at 13 ppm shifts towards 17 ppm. Motional narrowing as a result of inter-particle transfer becomes evident above 387 K, while at 440 K a very sharp signal at 16.9 ppm signifies crystallization of Li_7Si_3 . For the samples C-2000 and C-3000 analogous behavior can be noted, except that the crystallization of Li_7Si_3 occurs already at lower temperatures near 405 K. In these three samples Li_7Si_3 is then observed in the room temperature spectra of the heated samples (Fig. 4).

Fig. 5 summarizes the ^{29}Si MAS-NMR spectra of undoped nano-Si powder and the heat treated lithiated nano-silicon samples. The sharp signal at -81 ppm indicates crystalline silicon [38,39], while the shoulder might be due to disordered surface species [39–41]. Unfortunately, the signal of crystalline silicon is characterized by an extremely long spin lattice relaxation time [38], precluding any quantitative measurement on anode materials. As expected the C-500 and C-1000 samples present only the signal due to crystalline silicon, as a ^{29}Si MAS-NMR signal is very hard to detect for $\text{Li}_{15}\text{Si}_4$ [16,25]. Observation of crystalline silicon on these heated samples supports the contention that at these low-lithiation stages, a significant amount of sample material must still be present in the form of unintercalated nano-silicon, as also indicated in Table 2. The heated treated samples with higher charging capacities indicate the crystallization of Li_7Si_3 at 327 ppm for C-1500, and at 338 and 336 ppm for both C-2000 and C-3000. Chemical shift differences of this magnitude have been reported previously for other Li_7Si_3 samples and may be attributed to deviations of the Li/Si ratios from the exact stoichiometric composition [25].

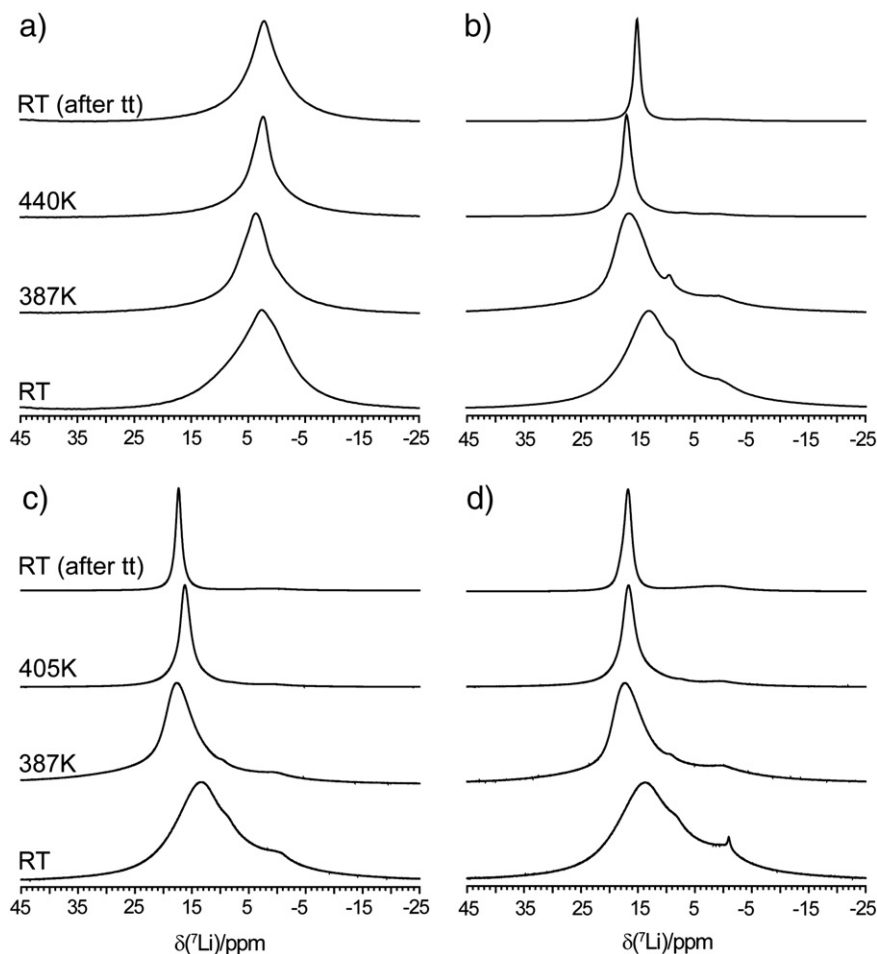


Fig. 4. Selected ^7Li MAS-NMR spectra acquired for the differently lithiated nano-Si anodes: a) C-1000, b) C-1500, c) C-2000 and d) C-3000. For each series, the uppermost spectrum was recorded several hours after the thermal treatment (tt).

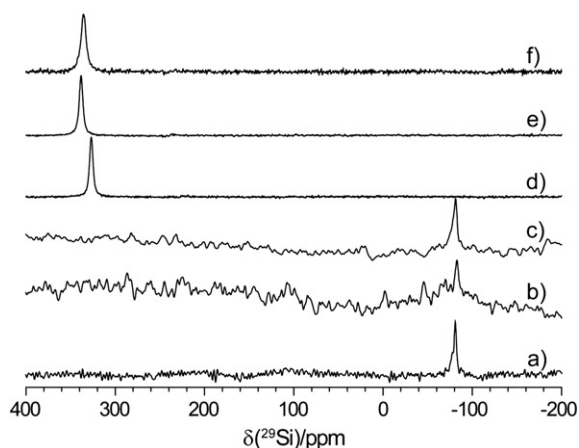


Fig. 5. ^{29}Si MAS-NMR spectra acquired for the pure nano-Si active material (a) and for the nano-Si anodes after thermal treatment: b) C-500, c) C-1000, d) C-1500, e) C-2000 and f) C-3000. Relaxation delays used were 80 s in case a) and 10 s in cases b–f.

We can compare the present results on the phase transition to crystalline Li_7Si_3 with those published by Dahn's group [30,31], obtained using DSC and XRD analyses. They observed the crystallization at about 448 K in anodes having an average stoichiometry of Li_1Si (954 mAh/g), Li_2Si (1908 mAh/g) and Li_3Si (2862 mAh/g). For Li_1Si and Li_2Si also a conversion of Li_7Si_3 into $\text{Li}_{12}\text{Si}_7$ at about 573 K was observed, and the material after the thermal treatment up to 773 K consisted mainly of $\text{Li}_{12}\text{Si}_7$, with a small amount of crystalline silicon. On the basis of the activation energies required, they concluded that the first event requires mainly Li diffusion, because the amorphous silicon framework is already mainly constituted of Si_2 dumbbells (as in Li_7Si_3), while the second requires diffusion of Si atoms and restructuring of the Si lattice, requiring a huge activation energy barrier [30,31]. In contrast to Dahn's result, no crystallization to Li_7Si_3 is observed at 1000 mAh/g in the present samples. This finding is consistent with Dahn's general conclusion, however, because in our 1000 mAh/g samples the amorphous lithium silicide phase mainly comprises monomeric silicon species. Crystallization of Li_7Si_3 is observed only for those materials (at a charging capacity of 1500 mAh/g and higher) which already contain dimeric silicon species in the amorphous state.

4. Conclusions

In summary, solid state ^6Li NMR can provide useful structural information about the lithiation process of nano-silicon particles during the first charging process. Our results indicate that at low charging capacities consistently "over-lithiated", $\text{Li}_{15}\text{Si}_4$ -like environments are formed, presumably in a kinetically controlled reaction, at the surfaces of the nano-Si particles. Formation of this environment is further promoted by the relatively low lithium mobility in this material. As lithiation progresses further towards the inside of the particles at higher charging capacities, new lithium environments comparable to those in the phases $\text{Li}_{13}\text{Si}_4$, Li_7Si_3 , and $\text{Li}_{12}\text{Si}_7$ are being formed, and above 2000 mAh/g the large majority of the Si nanoparticles is converted to lithium silicides. Lithiation of nano-Si doped with boron or phosphorus results in somewhat wider product distributions at certain charge capacities. In particular, phosphorus doping can reduce the irreversible capacity losses encountered at low charging capacities and – to some extent – diminish the preferential formation of $\text{Li}_{15}\text{Si}_4$ -like environments at these initial lithiation stages. The beneficial effect of P-doping at the initial stages of lithiation are in agreement with the explanation given by Long et al., that P-doping lowers the Li insertion potential

relative to that of the undoped material [26]. In addition, the higher electrical conductivity of P-doped nano-Si favors faster overall lithiation kinetics. Finally, our results indicate that high capacity losses are always observed, when $\text{Li}_{15}\text{Si}_4$ -like environments form the major part of the lithium inventory. The NMR results of the present study show that the occurrence of these environments can be diminished if P-doped materials are used. Furthermore, it is anticipated that for optimum reversibility and electrode stability it might be beneficial to limit cell cycling to the capacity region between 1500 and 3000 mAh/g. While this restriction seriously compromises the earlier promise of nano-silicon as an anode material for electromobility battery applications, it is worth pointing out that the remaining capacity is still at minimum four times larger than that available with graphite.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.ssi.2013.07.013>.

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